GOA Supermarket

Sales Forecasting & Time Series Analysis

Business Analytics Project

Project Overview

This project involved analyzing GOA Supermarket sales data to understand historical demand behavior and provide practical insights for inventory, replenishment, and promotional planning. The work focused on identifying sales trends, weekly seasonality, demand volatility, and differences between promotional and non-promotional periods.

Project status: Completed analysis of the available historical sales data at GOA Supermarket.

Business Objectives

- Understand how GOA Supermarket sales changed over time and identify recurring weekly patterns.
- Measure short-term sales volatility to highlight periods requiring closer inventory attention.
- Quantify the difference in average sales between promotional and non-promotional days.
- Translate observed demand patterns into practical recommendations for replenishment and promotional planning.
- Establish an analytical foundation for a future validated sales-forecasting model.

Analysis Performed

Analysis	Method	Business use
Historical sales trends	Daily sales time-series visualization	Understand demand movements, spikes and dips
Weekly seasonality	Additive decomposition with 7-day period	Identify recurring weekly demand patterns
Trend smoothing	30-day moving average	Separate short-term noise from broader sales direction
Promotion analysis	Average sales by promotion status + percentage difference	Support evidence-based promotion and inventory decisions
Sales volatility	30-day rolling standard deviation	Monitor periods of elevated demand variability

Quantified Finding: Promotions

The analysis compares average sales on promotional versus non-promotional days and calculates the percentage difference directly from the underlying dataset. The original chart shows the two averages at approximately 300 and 302 sales units, corresponding to roughly a 0.7% higher average on promotional days. The revised notebook now calculates and prints the exact values and percentage from the source data so the CV figure can be reported with appropriate precision.

Interpretation: The comparison is descriptive. A higher average on promotion days does not, by itself, establish that promotions caused the increase.

Business Insights and Recommendations

- Use recurring weekly demand patterns when reviewing replenishment requirements and operational planning.
- Use the 30-day moving average as a trend reference when assessing inventory needs.
- Evaluate promotional periods against baseline demand when deciding promotion timing and stock requirements.
- Pay closer attention to periods of elevated 30-day sales volatility because demand is less stable during those periods.

- Use the observed demand patterns as inputs to a future forecasting model that can be trained, validated, and evaluated with forecast-error metrics.

Forecasting Scope and Limitation

The source notebook created a 30-day future date range but did not contain a completed forecasting model or forecast-error metrics. The project materials therefore do not present fabricated forecast values or claim forecast accuracy. This leaves a clear next step: implement and validate a forecasting model using a time-based train/test split and metrics such as MAE, RMSE, or MAPE.

Project Outcome

The project converted historical GOA Supermarket sales data into measurable demand-pattern insights and practical recommendations for inventory alignment, replenishment planning, and promotional decision-making. The analysis supports a shift from reactive review of sales to more structured, data-driven business planning.